Automatic team-throughput forecasting from repository events

Operational model and evaluation protocol, 29 October 2026

Abstract

How many pull requests will a monitored team merge during the next complete calendar period? We specify an automatic forecast of this observable count, including requests created after issuance. The input is a paginated repository inventory with stable identities, merge timestamps and collection receipts. No labour diary, task classification, acceptance judgement or AI-use label is required. The operational reference is an empirical distribution over the supported completed-period history. It assumes that the selected history is informative about the next period, without asserting stationary queues, independent days or guaranteed prediction coverage. Current open-inventory predicates are descriptive; future bottlenecks are not predicted. On a predeclared 26-week public-reference comparison, a recent-window challenger has worse distributional loss than its expanding-history comparator and fails promotion. The operational reference is therefore retained, without a claim that this separate comparison validates its full-history deployment. The inspected user repositories have too little calendar history to establish predictive skill. Their daily cold-start forecast remains experimental; weekly and fortnightly outputs await the required history. The contribution is an inspectable operational model with its negative comparison and evidential limits retained.

1 A forecast that the recorded events can support

An empty pull-request queue does not imply that a team will merge nothing tomorrow. New requests can be opened and merged within the forecast period. Conversely, a large open queue does not imply a large forthcoming output: requests can wait, be abandoned, or remain outside the team's immediate priorities. A forecast of total merge activity must include future arrivals and distinguish calendar throughput from the resolution of today's backlog.

We choose a narrow operational question: how many distinct pull requests (PRs), visible through GitHub in a fixed set of repositories, will have a merge timestamp in a specified future interval? Selecting the repositories defines the monitored portfolio. It does not establish that all contributors belong to one team or that all of that team's work occurs there. Counts are sensitive to how work is split into PRs, automation and repository policy. We consequently retain the name *merge count* throughout the analysis.

The proposed workflow is automatic: collect the visible PR inventory, construct complete calendar counts, issue a distribution over a future count, and score the forecast after its horizon has elapsed. This is the recommended initial operational model. The companion labour-identification manuscript remains a separate measurement-research option for settings with additional labour observations. Its sharp bounds and sampling results are neither inputs to this forecast nor evidence of its accuracy.

2 Observation contract and calendar boundaries

Let $\mathcal{R}$ be a fixed set of stable repository identifiers. Deduplicate PR records by repository identity and PR identity, retaining the raw response, its content digest, collection time and pagination coverage. The official PR endpoint exposes the creation and merge timestamps needed for this count [4]. A lightweight inventory collector is sufficient; code diffs, workflow logs and personal activity labels are not required.

For a future half-open interval $I = [a, b)$, define

$$Y_I = \sum_{r \in \mathcal{R}} \sum_{i \in \mathcal{P}_r} \mathbf{1}\{a \le m_i < b\}, \tag{1}$$

where $\mathcal{P}_r$ denotes the retained visible PR identities in repository r and m_i is the merge timestamp, when one exists. An unmerged closure contributes zero. A close–reopen–merge history contributes one at its eventual merge. Requests created after a are included when they merge before b. A timestamp equal to b belongs to the next interval.

This is a visible-inventory estimand. Successful pagination establishes what the current access scope returned; it does not prove that deleted, transferred or inaccessible historical records never existed. An incomplete source response cannot establish a zero count. Every included portfolio interval must lie within the declared observation frame and after all selected repositories exist. Discard the initial partial period and the current incomplete period. A day before a repository existed is not a zero-activity observation for that repository.

The implemented source contract binds each throughput-collection summary to the PR records in the same retained manifest. It records collection start and end times, required sources and their coverage. For the selected repositories, set the complete-through cutoff conservatively to the earliest start time among their latest usable collections. Only training bins ending at or before that cutoff are eligible. This prevents a stale inventory, or a paginated collection crossing midnight, from manufacturing additional zero days. Record the complete-through cutoff separately from snapshot and issuance time. Current inventory remains a scan over a time span, not an atomic historical snapshot.

Three clocks. Retain the actual collection and forecast issuance times, the event cutoff used for training, and the target's start and end. A forecast issued during a UTC day targets the next complete UTC day, not the already partly observed day. Weekly periods start on Monday at 00:00 UTC. Training intervals end at or before the event cutoff; the prospective target starts no earlier than issuance. These choices leave a possible gap between the most recent completed training interval and the future target, which must remain visible.

Two evidence modes. *Retrospective reconstruction* uses timestamps in today's visible inventory to form earlier counts. It assumes that those timestamps faithfully describe the earlier events; it does not establish what an observer actually knew then. *Prospective evaluation* uses source receipts and forecast artefacts retained before the target starts by the automatic workflow. Save the repository scope, source or snapshot digest, model version, fitted intervals, target interval and issued probability distribution. These retained timestamps and content hashes support reproducibility; they are not independent trusted-time certification. Re-running today's software with an old cutoff does not turn reconstruction into prospective evidence. Mutable current draft status, head commit, review state or change size must not enter historical count predictors.

3 The empirical reference and recent-window challenger

Suppose $x_1, \ldots, x_n$ are the selected preceding complete-period portfolio counts. The default reference uses all supported complete periods in the declared portfolio frame. The issued probability mass function and cumulative distribution function are

$$\widehat{p}(k) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}\{x_j = k\}, \qquad \widehat{F}(k) = \frac{1}{n}\sum_{j=1}^{n} \mathbf{1}\{x_j \le k\}. \tag{2}$$

Repeated counts retain their frequency, including genuine zero periods. For a portfolio, sum the synchronized repository counts within each historical interval before applying (2). Do not convolve separately fitted repository distributions: that would impose a dependence structure absent from the observations.

The calendar and minimum-support requirements below were fixed before evaluating model scores. They are design choices, not sample-size theorems. The recent-window challenger was tested with a fixed history cap and was not promoted after failing the primary score criterion. The operational reference retains the expanding-history rule.

Target	Historical observations	Availability rule
Next complete UTC day	All supported complete daily counts	At least four complete days.
Next complete UTC week	All supported complete weekly counts	At least four complete weeks.
Next 14-day interval	All supported disjoint two-week totals	At least four two-week totals: eight complete weeks. No daily or weekly convolution.

The recent-window challenger instead uses at most eight daily or weekly observations, or four fortnightly observations, with the same four-observation minimum. The daily target is a cold-start amendment for repositories too young to support the frozen weekly benchmark; it does not replace that benchmark. Multiplying a daily forecast by seven, or convolving seven copies of its distribution, does not supply a weekly forecast under this protocol. Fourteen-day evaluation uses every other Monday, with parity anchored to 5 January 1970, so adjacent scored target intervals do not overlap. The method and model version are retained explicitly in each forecast.

Report the empirical mean as expected merge count and a predictive median as the point forecast evaluated by absolute error. Report the inverse-CDF 10th and 90th percentiles, training size and actual probability mass enclosed by those inclusive endpoints. For eight distinct observations, those percentiles are the sample minimum and maximum, so the interval contains all mass in the issued empirical distribution. Calling it an exactly 80% probability interval would obscure this discreteness. Its future coverage must be measured.

The historical maximum is not a bound on future activity. Equation (2) assigns zero mass to unobserved counts, including an unprecedented burst. Likewise, four zero days yield a point mass at zero without establishing that future activity is deterministic. The report must show the limited support on which the forecast rests.

3.1 What is assumed, and what is not

Computing (2) requires no Poisson arrival process, independent PR completion, independent calendar periods, proportional hazards or stationary queue equilibrium. Using it as a forecast nevertheless proposes $\mathbb{P}(Y_{\text{next}} \le k \mid \mathcal{H}) \approx \widehat{F}(k)$, where $\mathcal{H}$ is the recorded history. The

proposed approximation, rather than an equality theorem, is what future scores test. It assumes that the selected historical outcomes are informative about the future interval. That transport assumption can fail after a launch burst, a change in working pattern, a repository migration or a shift in PR granularity. The expanding reference can retain obsolete activity; the recent challenger can discard informative history. Neither has a universal advantage. Four to eight days offer little evidence about day-of-week effects and almost none about longer cycles.

No informative upper bound on future merges follows from an unrestricted observed past alone. Two continuations can share every recorded event through the cutoff and then produce zero or many merges. The method therefore supplies a deliberately simple, falsifiable forecast, not an assumption-free consequence of event accounting. More elaborate models should earn their additional data and assumptions through better future predictions on the same target.

4 Queue pressure without an unsupported capacity claim

A fresh inventory can report the last-observed number of open PRs, their creation ages and the oldest recorded items. The PR list also supports counts of open drafts, open requests with a nonempty requested-reviewer or requested-team list, and open requests at least seven days old. The seven-day threshold is a fixed descriptive choice. Optional fields that were not supplied remain unavailable; a known empty open inventory has zero items satisfying these predicates. Compute age at the source observation time and report the collection span and its age at forecast issuance. Creation age is elapsed time since opening, not uninterrupted waiting for review and not active processing time. Submitted-review status and CI outcomes remain uncollected by this profile. These predicates can overlap; they are not disjoint stages in a serial workflow and never enter retrospective count features.

For a complete history of state transitions, let B be open inventory, A new openings, R reopenings, M merges and C closures without merge. On an interval I,

$$B_{\text{end}} = B_{\text{start}} + A_I + R_I - M_I - C_I. \tag{3}$$

Classify a merge-induced closure once, in M_I, rather than subtracting it again in C_I. Final `closed_at` values do not reconstruct repeated close and reopen episodes. GitHub exposes relevant timeline events, but the lightweight throughput profile does not require or claim to reconstruct that history [3].

Equation (3) checks retrospective reconciliation when its events are observed. It does not forecast its future terms. Little's finite-window analysis makes the same distinction between sample-path measurement and forecasting: initial inventory and the time accumulated within the observation window require explicit treatment [7, Sections 2.1–2.3]. Dividing today's backlog by mean latency among completed PRs does not thereby estimate next week's throughput or the effect of adding a reviewer.

The recent *Queue & AI* model illustrates how service-time variability and rework can affect queueing delays. Its inputs include human-attention service requirements, effective reviewer capacity and error/rework quantities; its main calculation uses a stable single-server queue with Poisson arrivals, and its alternative invokes a queue approximation [1, Sections 2.1, 3 and 4.2]. PR timestamps do not supply those inputs. This mechanism is useful context for a possible future structural study, but it cannot be treated as an automatically calibrated model of the monitored repositories.

Similarly, an initial CI failure is an observable workflow outcome, not a measure of product correctness. A later successful rerun must not overwrite that initial outcome. The existing collector can retain run attempts separately, and the official API distinguishes them [2]. Missing, canceled and skipped runs require their own treatment. Reopening a PR also does

not establish that a defect was found. These signals may become forecast predictors or additional endpoints; they do not identify the cause of a throughput change or the effect of a staffing intervention.

The present implementation forecasts neither future queue pressure nor future bottlenecks. That part of the operational objective remains unsupported by the available history. A bounded follow-on would retain regular automatic snapshots, define a future observable pressure count, and test its forecasts on subsequent mature snapshots. Even a successful forecast of that count would not establish which resource constraint causes it or the effect of changing staffing. No such extension is claimed here.

5 A test that can reject the model

At each historical origin, train only on completed earlier intervals and evaluate on a fully elapsed target with adequate source coverage. A future merge may be an outcome but cannot become a training feature. Retain unsupported, incomplete and immature origins in the accounting of unavailable results. Do not score early-known positive outcomes while omitting unresolved horizons. The retrospective and prospective evidence modes remain separate.

The automatic archive scorer accepts original command-issued bundles containing the source snapshot, prepared history, method and forecast. It checks the declared creation time against the source cut and target start, verifies the source digest, reconstructs the history, and recomputes the forecast rather than trusting supplied probabilities. Offline reconstruction outputs are not issuance receipts. A later collection must cover the same numeric repository scope and have a conservative complete-through cutoff at or beyond the target end before the outcome is scored. Earlier horizons remain pending; incomplete evidence and malformed archives remain visible as unavailable results or errors. The archive is not rewritten during scoring. These checks bind the retained recipe and declared chronology, within the timestamp limitation in Section 2.

The primary loss is the discrete ranked probability score (RPS), or CRPS restricted to nonnegative integers,

$$L(F, y) = \sum_{k=0}^{\infty} \big(F(k) - \mathbf{1}\{y \le k\}\big)^2. \tag{4}$$

For a finite empirical distribution, an independent computational check is

$$L(\widehat{F}, y) = \frac{1}{n}\sum_{i=1}^{n} |x_i - y| - \frac{1}{2n^2}\sum_{i=1}^{n}\sum_{j=1}^{n} |x_i - x_j|. \tag{5}$$

This uses the nonnegative loss orientation of the standard CRPS identity [5, Section 4.2]. The formal independent draws from F in that identity are computational draws from the issued distribution; they do not assert independence of the historical counts. A point-mass forecast reduces to absolute error, allowing direct comparison with deterministic baselines. A previously unseen large count receives a finite penalty.

The score rewards the complete distribution. Narrowness alone is not calibration; calibration concerns agreement with outcomes [6]. Report mean RPS, paired losses against each fixed comparator, mean absolute error of the predictive median, empirical-quantile width, achieved target coverage, training support, scored origins and missing or immature denominators. The predictive mean is reported separately from the median used for MAE.

The frozen weekly comparators are an expanding-history empirical distribution, a zero point mass, the previous-week point mass, and a Poisson distribution whose mean is the preceding four weekly counts' mean. The Poisson model is a restrictive benchmark, not an assumption that these data are Poisson. Apply the analogous frozen comparators to the

experimental daily target. The recent and expanding empirical models intentionally coincide until observations age out of the recent window. Select neither the window nor the best comparator anew after inspecting each outcome.

For a bounded public-reference exercise, the repository `cpmpy/cpmpy` and a 78-week window were chosen before inspecting model scores. Its first 52 complete UTC weeks initialize the history; the following 26 weeks provide chronological targets, with each subsequent forecast allowed to use earlier observed evaluation weeks. The expanding comparator expands from that declared 52-week initial history. This tests a fixed recipe in one reference population; it does not supply an independent test of every possible model or evidence of transfer to the user's repositories.

The operational default subsequently retained is the full-supported-history reference, whereas the study's expanding comparator uses only the declared 52–77-week training frame. These are different observation frames. The failed promotion of the recent challenger does not establish the predictive accuracy or global superiority of the deployed full-history reference.

A provisional practical-improvement screen is at least 5% lower mean RPS than the expanding empirical comparator, without material MAE regression. The threshold is an engineering choice. If the comparator has zero loss, percentage improvement is unavailable. A confident superiority claim also needs adequate calendar replication and a predeclared dependence-aware uncertainty analysis with a positive lower improvement bound. Nonoverlapping periods can still be serially dependent, and two repositories are not automatically two independent replications. Resampling a few days cannot manufacture the missing evidence.

Implementation checks. Changing or appending events after a cutoff must leave its forecast unchanged. Exact boundary timestamps must enter the correct half-open interval. Duplicate source versions must leave counts unchanged. Complete zero periods must survive; incomplete source pages must not become zeros. New requests merged within the target must count. Current mutable fields must not alter historical count forecasts. Independent evaluation of (4) and (5) checks numerical correctness, not predictive skill.

6 Current evidence and operational use

The inspected user repositories have only several days of history. That is enough to exercise automatic inventory extraction and, when four full days exist, issue an experimental daily distribution. It is not enough to validate weekly throughput, estimate seasonality, or establish comparative predictive skill. With five complete daily observations the expanding reference and recent challenger issue the same distribution. Weekly and fortnightly unavailability is an informative result of the observation contract.

Repository	Created (UTC)	Visible PRs	Merged PRs
emmett08/eal-mcp	2026-10-20	33	33
emmett08/engarg	2026-10-23	45	45

The table and distribution below are offline reconstructions from a retained visible-inventory snapshot, not evidence that a forecast was actually issued at that snapshot's declared cut. The inventory includes all visible PRs, including merges outside the fitted portfolio period. Its conservative complete-through cutoff is 2026-10-29T22:00:10.900156728Z. The last-observed open inventory is 0; this does not constrain future arrivals. The expanding reference uses 5 complete daily counts: $(17, 11, 13, 12, 8)$, ordered by date. The replayed prediction for [2026-10-30, 2026-10-31) UTC has mean 12.2, median 12, and empirical 10th–90th percentile range 8–17. That inclusive range contains 100% of the reported empirical mass, with no

guaranteed future coverage. The target is future relative to the declared snapshot cut; this replay is not a mature forecast score. The weekly report contains 0 complete historical weekly bins and is unavailable. These feasibility outputs establish automatic operation, not forecast accuracy.

A separate original `forecast-team` bundle is retained in `examples/issued-20261029.json.gz`. Its locally declared creation time is 2026-10-29 22:18:19 UTC, for the target [2026-10-30, 2026-10-31) UTC. Its source snapshot differs from the replay's, but its same five complete counts produce the same numerical distribution. The accompanying score is pending and contains no outcome or loss. This original command receipt is distinct from the offline table reconstruction; its timestamps remain declarations rather than independent trusted-time certification.

6.1 Predeclared public-reference comparison

The following values are generated from the fixed 26 chronological targets in `cpmpy/cpmpy`. MAE uses the predictive median. The empirical percentile range is descriptive; its mean width and observed coverage are reported separately from its requested quantiles.

Method	Targets	Mean RPS	MAE	Mean width	Coverage
Recent empirical	26	2.526	3.038	13.12	80.8%
Expanding empirical	26	2.275	3.538	10.15	92.3%
Zero	26	4.615	4.615	0.00	15.4%
Previous period	26	4.500	4.500	0.00	15.4%
Poisson, prior-four mean	26	2.766	3.731	5.69	42.3%

The recent empirical model's mean RPS is 11.03% higher than the expanding empirical comparator's on this fixed reference window. This signed result is retained regardless of whether it favours the recent-window model. No dependence-adjusted confidence interval or superiority claim is supplied, and no result is transferred to the user's repositories. The recent-window model does not meet the predeclared 5% distributional-improvement screen on this reference window. Its point-error result cannot override that primary criterion. The complete report retains target dates, outcomes and paired losses for every method. All results are retrospective reconstructions from the retained visible inventory.

Any longer-lived public repository used to exercise the evaluation machinery is a separate reference population. Successful collection, score calculation or even forecast improvement there does not establish accuracy for the user's repositories. The immediate output is an inspectable forecast with its event scope, clocks, historical support and provisional status. Keeping those forecasts unchanged and scoring their mature outcomes is the next source of predictive evidence.

The operational contribution is a small observation-to-forecast contract that can be implemented without asking people to classify their work. It supports a useful count forecast when history is informative and makes failure visible when it is not. Genuine capacity bottlenecks, labour productivity, correctness and the causal effects of AI or staffing remain different questions requiring additional observations or justified assumptions.

References

[1] Silvia Bartolucci and Pierpaolo Vivo. Queue & AI: When faster tasks slow down the workflow. arXiv:2605.27202v1, 2026. URL https://arxiv.org/html/2605.27202v1. Version 1, 26 May 2026; accessed 29 October 2026.

[2] GitHub. REST API endpoints for workflow runs: Get a workflow run attempt, 2026. URL https://docs.github.com/en/rest/actions/workflow-runs#get-a-workflow-run-attempt. Official documentation, accessed 29 October 2026.

[3] GitHub. Issue event types, 2026. URL https://docs.github.com/en/rest/using-the-rest-api/issue-event-types. Official documentation, accessed 29 October 2026.

[4] GitHub. REST API endpoints for pull requests, 2026. URL https://docs.github.com/en/rest/pulls/pulls. Official documentation, accessed 29 October 2026.

[5] Tilmann Gneiting and Adrian E. Raftery. Strictly proper scoring rules, prediction, and estimation. *Journal of the American Statistical Association*, 102(477):359–378, 2007. doi: 10.1198/016214506000001437. URL https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jasa.pdf.

[6] Tilmann Gneiting, Fadoua Balabdaoui, and Adrian E. Raftery. Probabilistic forecasts, calibration and sharpness. *Journal of the Royal Statistical Society: Series B (Statistical Methodology)*, 69(2):243–268, 2007. doi: 10.1111/j.1467-9868.2007.00587.x. URL https://sites.stat.washington.edu/raftery/Research/PDF/Gneiting2007jrssb.pdf.

[7] John D. C. Little. OR FORUM—Little's Law as Viewed on Its 50th Anniversary. *Operations Research*, 59(3):536–549, 2011. doi: 10.1287/opre.1110.0940. URL https://people.cs.umass.edu/~emery/classes/cmpsci691st/readings/OS/Littles-Law-50-Years-Later.pdf.